

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable

example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I B . M a h e s h w a r (1 9 2 4 1 1 1 9 5) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Maheshwar

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
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Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application	20	Correctly	Applies	Limited or	No or

of Knowledge		applies concepts to solve the question/problem	concepts with minor mistakes	partially correct application	incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response

1) Responsive web Design and CSS Layouts

Responsive web design is a web development approach that ensures a website adapts to different screen sizes such as mobile phones, tablets, and desktops. Its importance lies in providing a consistent user experience, improving accessibility, and reducing the need for separate websites for different devices.

Key CSS Techniques:

- fluid layouts using percentage widths
- flexible images using `max-width: 100%`
- Media queries for different screen sizes
- CSS flexbox and grid for adaptive layouts

flexbox example:

```
.container {  
  display: flex;  
  flex-wrap: wrap;  
}
```

```

.item {
  flex: 1;
  min-width: 200px;
}

```

Improvements:

- use responsive typography & optimize images for faster loading.
- ensure touch-friendly buttons on mobile

2) Client-side validation using javascript
 client-side validation is the process of checking user input in browser before submitting data to the server. its purpose is to improve user experience, reduce server loads, prevent invalid data entry.

JavaScript Event Handling:
 Events such as submit, keyup, blur triggered validation functions

Role of Regular Expressions:
 Regular Expression and patterns used to validate input formats.

```

<form onsubmit="return validateMail()">
  <input type="text" id="email">
  <input type="submit">
</form>
<script>

```

```

function validate Email () {
    let email = document.getElementById("email").value;
    let pattern = /^[^ \s@]+@^[^ \s@]+\.[^ \s@+$/;
    return pattern.test(email);
}
</script>

```

Improvements:

- Display clear error messages.
- validate data on both client and server sides

3) CSS Box Model, positioning, and media queries

The CSS Box model describes how elements are displayed on a webpage. it consist of:

1. Content - actual text or image
2. padding - space around content
3. Border - surrounds padding & content
4. margin - space outside the border

CSS positioning Techniques:

- Static - default position
- Relative - positioned relative to its location
- absolute - positioned relative to the nearest positioned ancestor
- Fixed - remains fixed on the screen
- Sticky - switches between the relative and fixed.

Causes of Layout breaking;

- Fixed widths
- Edge images
- Improper positioning
- Lack of responsive design

media query example:

```
@media (max-width: 768px) {  
    .container {  
        flex-direction: column;  
    }  
}
```

Best practices:

- Use flexible layouts
- Apply media queries for various screen sizes.
- Test on multiple devices & browsers.

4) DOM and Dynamic content updates:

The Document object model (DOM) is a programming interface that represents an HTML document as a tree structure. JavaScript uses the DOM to access and modify web page content dynamically without

Reloading the page.

Role of DOM in web Development:

- Access HTML elements.
- Modify content, styles, and attributes.
- Respond to user interactions.

Example:

```
<P id = "demo" > welcome </P>  
<button onclick = "changeText()" > click </button>  
<script>
```

```
function changeText() {
```

```
    document.getElementById("demo").innerHTML =  
        "Content updated!";
```

```
}
```

```
</script>
```

When the button is clicked, the paragraph's text changes without refreshing page

Improvements:

- minimize unnecessary DOM manipulations
- use event delegation for better performance
- update only required elements to improve user experience